# Towards Intent-based Network Management for the 6G System adopting Multimodal Generative AI

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***Abstract*—The emerging concept of delivering Network-as-a-Service (NaaS) foresees the deployment and reconfiguration of the next-generation networks, such as 6G, in a dynamic and elastic manner, tailored to the respective stakeholder's intention. Taking this into account, the efficient management and orchestration of both telecommunication and computational resources across the network domains, i.e. access, transport and core presents a considerable challenge, even for network experts. To tackle this complexity, this paper explores the implementation of an intent-based management framework. The framework receives a high-level description of the desired network capabilities along with supplementary files, e.g. deployment descriptors, and translates them into configuration files consumable by the network itself. In order to achieve this, the paper establishes a translation pipeline that leverages the employment of emerging multimodal generative artificial intelligence (GenAI) models, specifically Large Language Models (LLMs), and open industry-ready standard templates. The adoption of those two emerging technologies offers high dynamicity on the interpretation process of the user's intent, while ensuring that its outcome is compatible with every orchestrator or next-generation Operating Support System (Next-gen OSS) that adheres to those standards.**

***Keywords*—Intent-Based Networking, Large Language Models, Multimodal Generative Artificial Intelligence, Orchestration, Standards**

## I. Introduction

Over the past few decades, mobile networks have become an integral part of our daily lives, revolutionizing how we communicate, work, and interact with the world around us. However, it is in the last decade that we have witnessed an exponential surge in their presence across various industries and everyday scenarios. This widespread adoption of new technologies has not only transformed the way we connect but has also unearthed a lot of new challenges that must be addressed by the architects of next-generation networks.

As we navigate this rapidly evolving landscape, it becomes increasingly evident that the future of mobile networks depends on their ability to adapt and evolve. The cornerstone of this adaptation is the necessity for the said networks to be scalable and reproducible, ensuring their resilience and reliability amidst growing demands and technological advancements. Moreover, as mobile networks become more ubiquitous, the need for customization and programmability becomes a matter of major importance, particularly in catering to the diverse requirements of different verticals.

Furthermore, it is essential to comprehend that the network configurations required by various verticals are dynamic and may vary over time and context. Therefore, the ability to offer the Network-as-a-Service (NaaS), which encapsulates the necessary abstract to address this issue, emerges as a vital component in ensuring the flexibility and adaptability of future networks. However, for network operators to effectively deliver NaaS and manage the inherent complexities of scalability, there arises a pressing need for the enhancement of the existing orchestration platforms. These next-generation orchestrators must possess the capability to orchestrate end-to-end deployments, focusing on efficient network management, energy consumption and seamless service delivery, while also actively promoting the digital inclusion of non-technical stakeholders.

Until recently, the convergence of various networks and their seamless interoperability have been a focal point for both academic and industrial standardization bodies, both represented by high technical expertise. Efforts have been made to establish unified communication protocols and facilitate third-party interworking and inevitably the orchestration across diverse network infrastructures. At the very center of this initiative stands the 3rd Generation Partnership Project (3GPP), which serves as the principal architect of mobile networks. In parallel, industry organizations such as the TeleManagement Forum (TMF) and the GSM Association (GSMA) have emerged as key players in bridging the gap between industry requirements and networking components.

On the other hand, amidst these ongoing efforts, another area in networking has rapidly gained momentum. The eruptive advancements of mobile networks and their widespread adoption have created an influx of users, not necessarily fully competent in network intricacies. The aforementioned users seek advanced services, disregarding their realization inside the network. This has accelerated the need for user-friendly descriptions of their requirements, easily translatable to network consumable configurations. An endeavor to address this need has been the conception of Intent-Based Networking (IBN). While the IBN concept has been around already for some time, the recent ascendancy of Large Language Models (LLMs), Generative Artificial Intelligence (GenAI) and Multimodal Artificial Intelligence (AI) has made the prospect of properly realizing it more feasible than ever before, bringing yet a new era in network architecture and management.

To that extent, this work proposes a framework that capitalizes on the capabilities of cutting-edge AI and Machine Learning (ML) tools, along with widely used standards-based interfaces, to deliver NaaS in the most digitally inclusive manner as possible. Specifically, this framework supports:

- incorporation of stakeholders' intent through a User Interface (UI)
- context distillation from high-level descriptions and files by employing Multimodal GenAI models
- translation of the captured intent into machine consumable standardized interfaces

For that matter, the rest of the paper is structured as follows: Section II describes the related work regarding the standardization activities on network models and interfaces, along with IBN. Section III presents the proposed framework in detail. In Section IV, the solution's usefulness is validated through an indicative proof of concept (PoC), encompassing a text-based intent realization at this stage. Finally, Section V comprises the conclusion and future work.

## II. Related Work

### A. Intent-Based Networking

Intent can be defined as a set of operational objectives that a network is expected to achieve and outcomes that it is supposed to deliver, introduced in a declarative manner without specifying the implementation details. IBN aims towards networks that are fundamentally simpler to manage and operate, only requiring minimal intervention from the user [1].

IBN has gathered significant attention in different aspects, including conceptual frameworks, architectural designs, practical implementations, and applications of AI [2]. Organizations such as the Linux Foundation Networking (LFN) and the European Telecommunications Standards Institute (ETSI) are actively involved in defining [3], standardizing [4], and developing intent-based components for their orchestration frameworks [5].

IBN also comprises a wide range of functionalities that can be clustered into two main categories: Intent Fulfillment and Intent Assurance. Fulfillment involves intent ingestion, translation, and orchestration, focusing on providing the necessary functions and interfaces for the intent to be conveyed to the network and implemented (the work performed in our paper solely explores this aspect of IBN). Complementary to the Fulfilment, Assurance deals with functions and interfaces allowing users to verify that the network indeed meets the specified targets set by the Fulfillment [2], [6].

Furthermore, several initiatives have been undertaken in the areas of intent ingestion and translation. An initial version of this implementation model was executed in [7] to correlate user-provided keywords with keywords exploitable by a software-defined network (SDN) controller. In [8], the authors developed an intent language, which was translated into P4 templates to configure the network. Following a similar approach, in [9], the intent was translated into Network Service Descriptors (NSDs) by employing generative adversarial neural networks (GANs). More recently, with the rise of LLMs, more sophisticated use cases of IBN tend to emerge. For instance, in [10], a GenAI model was employed to provide policies for the network to consume. However, to the best of our knowledge, there has not been any stable implementation specifically targeting the adoption of the standards established by the standards development organizations (SDOs), something that serves as the primary motivation behind this paper.

In terms of relevant research projects in the field of IBN, notable investigation has been conducted in the European Union (EU) project HEXA-X-II. It envisions the design of an end-to-end (E2E) system design, integrating all the interacting enablers, and the respective platform that will enable service delivery for 6G networks [11]. The project's ambition specifically for IBN is to serve as a gateway to users with no prior knowledge of controlling or utilizing the resources beneath the application layer. To achieve that, AI techniques will be used to enable a form of "self-learning" about how to accurately understand and translate the customers' intents into precise management and orchestration activities [12].

### B. Network-as-a-Service

The introduction of IBN in sixth-generation (6G) wireless networks aims to address the lack of efficiency, flexibility, and security issues commonly encountered in traditional networks. IBN plays a key role in translating users' business intentions into network configuration, operational, and maintenance strategies, making them essential for the design of AI-enabled 6G networks. However, a strong prerequisite in order to meaningfully support IBN is the programmability of the network, both for deployment and configuration, often referred to as NaaS.

Considering NaaS, GSMA, TMF, CAMARA Project [13], and Linux Foundation along with twenty-five operators recently introduced GSMA Open Gateway [14]. This initiative represents the future paradigm shift whereby network operators expose and monetize network capabilities to third-party application service providers and developers, through Application Programmable Interfaces (APIs) in a programmatic manner. Since its launch, the main focus of the initiative was drawn towards the exposure layer to third-party customers, directly or through aggregators, via the Service and the Service Management APIs. Although the mutual development of these APIs among the operators and the third-party stakeholders facilitated the well-sought abstraction of the network, skyrocketing their adoption and usage, the need for the transversal (non-service specific) functionality that is required to make a product out of the Open Gateway services is still apparent.

The latter is captured by the initiative's Operate APIs offering the Operation, Administration and Maintenance (OAM) functionality, e.g. service fulfillment, supervision, and assurance, on top of which aggregators offer NaaS to the customers. These APIs are commonly utilized by the operator's IT stacks, including Operation and Business Support Systems, widely implementing TM Forum's reference digital architecture employing over 70 Open APIs [15]. To sum up, a contemporary platform would need to i) capture the high-level business requirements of the customer within the networking context domain and ii) translate them into a static set of offered network slices, signing the corresponding Service Level Agreement (SLA) with the customer. Therefore, the platform would define the service's end state and the respective compliance following best practices.

While the initiative is relatively new, some subsets of its components have been implemented by various standalone projects. One of those efforts comes from the ETSI Software Development Group for OpenSlice (SDG OSL) which develops an open-source service-based Operations Support System (OSS) to deliver Network Slice as-a-Service (NSaaS), implementing numerous TMF APIs towards that goal [16]. Furthermore, it incorporates an inherent mechanism to express the business requirements that GSMA dictates in GST [17], to formulate an SLA and consequently a network slice, into a TMF Service Specification, thus rendering them machine consumable, as described in [18]. For the aforementioned reasons, it becomes naturally apparent that OpenSlice could act on the Operate APIs plane of the Open Gateway initiative.

## III. Proposed Framework

In this section, we introduce a deployment framework that leverages the potential of state-of-the-art AI and ML tools, in addition to standardized APIs for intelligent network orchestration, as depicted in Fig. 1. Within this proposed framework, a user initiates the flow by expressing its high-level business intent/description and by providing supplementary files to request a network slice. These supplementary files could be deployment descriptors, such as HELM charts, NSDs, and archives. The request is processed by the Dialogue block, which parses the input for insights and then forwards it, along with the derived insights, to the LLM component for context analysis. Subsequently, a standardized network slice template (NEST) JSON [17] is generated by the LLM and then provided to the Dialogue block for the user to assess whether the suggested slice description accurately captures the intended requirements. If the user is not satisfied, this process is iterated; otherwise, the NEST JSON is handed over to the Translator component, where it is formatted in accordance with the Operate APIs of a Next-gen OSS, e.g. a TMF Service Order request. In succession, this request is forwarded to the OSS, where the Infrastructure Admin/Scheduler performs a feasibility check for the appealed network slice. Finally, following the administrator's approval, the OSS initiates the fulfillment of the captured Service Order by employing once again the respective lower-level Operate APIs, e.g. TMF Resource Ordering, to entrust the realization of the network slice to the Infrastructure.

### A. Dialogue Block

This element assumes the role of a facilitator within the framework by acting as the external access point for user interaction. As input, it receives a high-level description of the required network characteristics and any deployment-related files. The block analyzes them for essential keywords, and forwards these keywords, along with the original input, to the LLM block. Next, the Dialogue component recovers the resulting NEST JSON generated by the LLM, which can be optionally reviewed by the user. Subsequently, the generated NEST is passed on to the Translator block.

The purpose of the framework is to adopt the zero-touch concept. So, it is essential to minimize human intervention by configuring appropriately the Dialogue component to forward

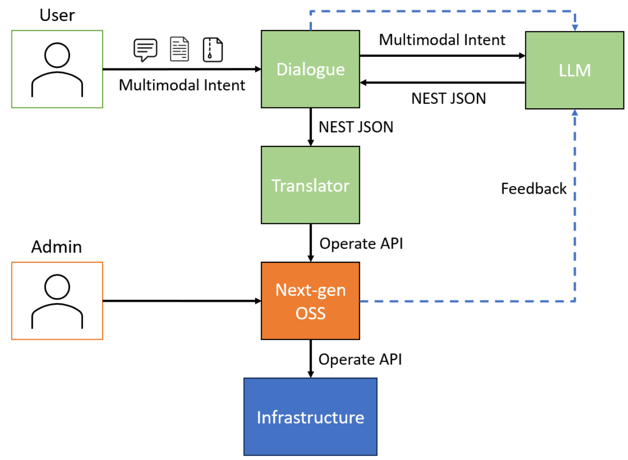


Fig. 1. Proposed framework's architecture

the NEST JSON without further review from the user's side, unless there is an explicit request for that. Nevertheless, if the user decides to actively interfere, the result of the review (acceptance, rejection, regeneration) is used by the Dialogue block to train the LLM during runtime.

### B. Large Language Model

GenAI and especially LLMs have attracted considerable attention in the natural language processing (NLP) domain because of their capability to generate coherent and relevant text. These models undergo training on extensive datasets, allowing them to comprehend and produce human-like responses to a range of queries. Constructed using deep learning techniques, particularly utilizing a type of neural network called a transformer, these LLMs have been proven to be highly effective in various NLP tasks. The transformer network serves as the central component of an LLM, consisting of multiple layers of self-attention mechanisms and feed-forward neural networks. When generating responses, the model employs the self-attention mechanism to assess the importance of different words or tokens in a sequence. The text input for LLMs is typically tokenized into smaller units, such as words or subwords, with each token associated with a unique numerical representation known as an embedding. Tokenization enables the model to systematically analyze text and capture relationships between tokens. During the inference stage, the model receives a sequence of tokens as input and generates a probability distribution for potential future tokens, ultimately selecting the token with the highest probability as its output.

To equip the general pretrained model with insights and knowledge in the networking domain, the use of few-shot learning [19] should be considered, given the shortage and depth insufficiency of the available datasets within that domain's context. This transfer learning approach leverages a minimal amount of information, including definitions, descriptions, examples, and behavior specifications (e.g. network engineer), as prompts [20]. These prompts are employed to fine-tune and customize the models, avoiding the need to adjust the model's entire parameter set. Moreover, this technique enables the addition of new tasks to the model without erasing or interfering with previous tasks. Consequently, it facilitates the model's ability to accumulate and cater to a diverse range of slice types or orchestration objectives.

Fig. 2 shows our proposed pattern of how a few-shot learning-based model can be trained. Following three main segments, each representing a different service type in the context of 5G and beyond networks. These service types are: Enhanced Mobile Broadband (eMBB), Ultra-Reliable and Low Latency Communications (URLLC), and Massive Machine Type Communications (mMTC). The eMBB segment includes attributes like ”Peak UL ($\approx$ 10 Gbps)”, ”Broadband”, ”Peak DL ($\approx$ 20 Gbps)”, and ”High Throughput”. The URLLC segment focuses on ”Mission-Critical” aspects with attributes like ”Reliability ($\geq$ 99.999)”, ”Low Latency ($\leq$ 5 ms)”, and ”Real-Time”. The mMTC segment is associated with the ”IoT” and includes attributes such as ”Density ($\geq$ 1 device/m)”, ”Massive Connectivity”, and ”Low Power”. This type of chart is used to illustrate the different use case categories that the aforementioned technology aims to address and is used for our training purposes, to explain the different patterns of the generated datasets.

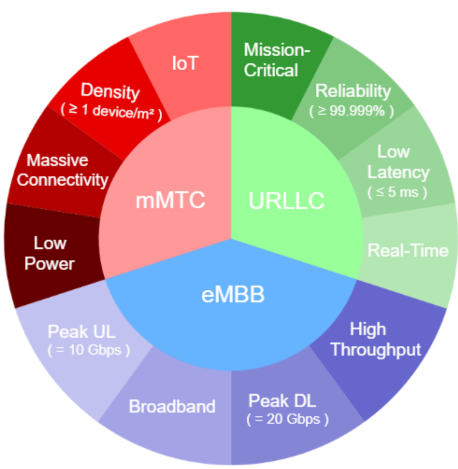


Fig. 2. Network knowledge insights to training data.

During the inference phase, the user's business intent can optionally be supplemented with the deployment descriptor files. This allows the LLM to access additional information for processing, resulting in more targeted outcomes for the NEST JSON that is provided to the Translator.

### C. Translator

As future networks diverge from the typical SLA enforcement track of policy-based static rules and pave towards more elastic, range value-based and dynamic SLAs, it is essential to rely on a communication model adequately flexible, but also standardized to avoid segmentation. Specifically, modern SLA components additionally comprise key value indicators (KVIs), e.g. energy efficiency and security aspects, along with the respective traditional key performance indicators (KPIs). Whereas the policy-based approach was sufficient from the standpoint of the legacy SLA, being a list of measurable KPIs, their contemporary counterparts include abstract non-quantifiable values, often deriving from non-native network context language, thus an alternative approach is required.

In this work, we attempt to leverage a well-established generic slice template, i.e. GST, to generate an SLA by solemnly residing on the business intent of the user for that matter, assisted by GenAI. Also, to ensure compliance with the Open Gateway platform we propose and implement a mechanism for transforming the produced SLA, or the NEST JSON, into a TMF-compliant model, namely the Service Order, that can be utilized directly by the respective TMF Operate API consumer. Specifically, the user's business intent is mapped to the output NEST JSON, comprising certain populated GST properties, which are then modeled as a TMF Service Order request with particular characteristics. The aforementioned process is illustrated in Fig. 3. Finally, the constructed Service Order is consumed by a Next-gen OSS, which supports the respective TMF Operate APIs, and undertakes its fulfillment.


Fig. 3. NEST JSON transformation to TMF Service Order

## IV. Proof of Concept

The presented PoC took place at the Patras5G testbed [21] which is equipped with competitive computational capabilities and a broad range of telecommunication devices, enabling the overall endeavor. The PoC materialized only the the text-based side of the proposed framework in order to explore its potential, paving the way on the same time to extend towards the support of multimodal intent recognition. Specifically, it leverages GenAI to transform an expression of high-level business requirements to the corresponding Operate API call, which is forwarded to a Next-Gen OSS to facilitate NaaS, offering the appropriate network slice.

Both the Dialogue and the Translator blocks were developed as microservices and deployed over Kubernetes. As for the LLM, at this stage, the framework leverages OpenAIs API to establish an HTTPS internet connection with the GPT-3.5-turbo model. The generated inference is fed to OpenSlice, serving as the Next-gen OSS which ultimately fulfills and delivers the originated network slice request.

### A. Training the LLM

The LLM was initially configured to function as a classifier, categorizing each intent description into one of the following groups following the proposed approach of Fig. 2. Upon deciding in which class the input data belongs, it forwarded that result and it was up to the Dialogue block to provide a static NEST JSON file as described by GSMA. Table I showcases the prompts along with the targeted classes that were provided as input to the LLM in order to perform the few-shot training stage.

TABLE I. Prompt Examples and Targeted Classes

| | |
|---|---|
| I want a network slice to support a massive IoT<br>I want a network slice to support a wide number of devices.<br>I want to support a large number of devices.<br>I want a network slice for industrial use. | mMTC |
| I want a network slice to support URLLC<br>I want a network slice to support low latency communication.<br>I want a network slice to support critical communication.<br>I want fast and reliable communication.<br>I want a network slice for XR. | URLLC |
| I want a network slice to support eMBB.<br>I want a network slice to support wide broadband communication.<br>I want a network slice with extended channel capacity.<br>I want a network slice for 4K videos.<br>I want heavy traffic communication. | eMBB |

However, in order to exploit the GenAI aspects to the finest, it was decided to set the model to operate in a more elastic and dynamic manner. This means that the model once again receives the intent description along with the insights from the Dialogue block, but it now provides the NEST JSON filled in with values of its choice.

The training process requires providing the model with a high-level description along with the annotation that the given description matches a specific slice type for the static configuration or entire NEST JSON files for the dynamic one. An example of such a JSON file that could describe the mMTC slice can be found below.

```
{
   "Availability":"99,9",
   "Slice quality of service":"9",
   "Supported device velocity":"1",
   "UE density":"1000"
}
```

## B. Inferencing: From user intent to Service Order

The flow is initiated as the user/customer provides its high-level business intent to the Dialogue block, which subsequently performs the matching to the applicable NEST JSON by employing the LLM. On that notion, several high-level business intent examples that lead to the corresponding mMTC, URLLC, and eMBB-based SLAs are transpired in Fig. 4, 5, and 6, respectively.

```
Assistant: How can I help you?
I am looking for a solution that can efficiently
manage a number of low power devices for industrial
applications that are scattered over a geographical
area.

Assistant: Massive IoT - mMTC
{
  "Availability":"99,9",
  "Slice quality of service":"9",
  "Supported device velocity":"2",
  "UE density":"10000"
}
```

Fig. 4. mMTC NEST generation

```
Assistant: How can I help you?
I am in need of a solution that supports high-speed
and highly reliable communication, specifically tai-
lored on a fast moving vehicle.

Assistant: Ultra-low latency communication - URLLC
{
  "Availability":"99,9",
  "Session and Service Continuity Support":"1",
  "Slice quality of service":"82",
  "Supported device velocity":"3"
}
```

Fig. 5. URLLC NEST generation

```
Assistant: How can I help you?
I am looking for a solution that enables the support of
4K video streaming over the public internet.

Assistant: Enhanced Mobile Broadband – eMBB
{
  "Availability":"99,999",
  "Session and Service Continuity Support":"SSC mode 1",
  "Data Network access":{ "Internet DNN":" Direct access
to the internet"},
  "Supported data network": "Internet DNN" ,
  "Slice quality of service": "9"
}
```

Fig. 6. eMBB NEST generation

Next, the generated SLA, expressed in the form of a NEST JSON, is fed to the Translator block which ultimately transforms it to a Next-Gen OSS-ready Operate API request, as detailed in Section III-B. It should be pointed out that this step does not intervene with the generated NEST JSON, but it solemnly transforms it to encapsulate an Operate API-native context that can be comprehended by the next-gen OSS of the architecture, i.e. OpenSlice. The particular Operate API that was appointed for this interaction is the TMF641 Service Order API, which signifies the transition from the business to the service domain.

## C. Service Order fulfilment

The final step comprises the capture of the aforementioned Operate API request from OpenSlice, the Next-Gen OSS in our implementation. Since the request's payload, based on the content of Fig. 4 is already in a native OSS context, OpenSlice can resolve it directly as appeared in the corresponding Service Order management User Interface (UI) in Fig. 7.

Henceforth, the Infrastructure Administrator can carry out a feasibility check on the requested network slice and, once successful, perform the acknowledgment of the Service Order. This acknowledgment process has always been rather typical of any system, regardless of its operational domain, and we believe that it could also be revolutionized, thus omitted, by employing promising 6G techniques, as introduced in the next section. Last but not least, following the Service Order acknowledgment, the request is delegated for fulfillment to the integral component of Openslice, namely the Service Orchestrator. At this stage, another set of Operate APIs is employed to validate the transition from the service to a lower

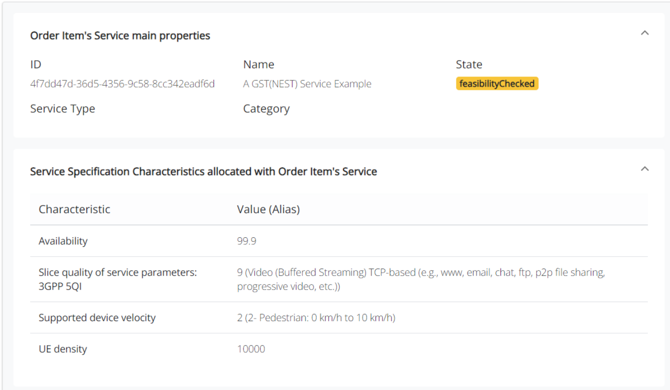


Fig. 7. Service Order management UI

domain, i.e. infrastructure/resources, and the actual realization of the high-level intent to a tangible network deployment, which is out of the scope of this paper.

## V. Conclusion

This work presented and demonstrated that through the employment of the appropriate generative AI models in conjunction with the adoption of standardized network APIs, the realization of NaaS was effectively facilitated and successfully implemented. This achievement stems from enabling users to express their network requirements in an intuitive and agile manner, thereby bridging the gap with the underlying traditional network domain, i.e. orchestrators and infrastructure.

As future work, the authors of the paper envision the complete expansion towards the encapsulation of the described multimodal intent recognition which would cognitively support the provision of archive files, along with the high-level business intent. Apart from the fact that this could magnify the accuracy of delivering NaaS, it would also merge the network preparation and application deployment in a single seamless step, further enhancing the user's experience and digital inclusion.

Furthermore, the deployment of a local LLM with fewer parameters is also targeted so as to cover the efficient functionality in resource-constrained computational environments. This model will be trained to narrow down and focus on specific network aspects rather than general information, ensuring that the reduction in parameters does not compromise its programmability and overall capabilities. NaaS deployment will thus be enhanced since this directed LLM will be able to generalize the NESTs and hence provide the requested services in a fully dynamic manner. Additionally, the training phase of the LLM will be expanded by utilizing the decision of the infrastructure administrator regarding the acceptance of the Service Order, in order to better adapt to the infrastructure's characteristics.

Ultimately, a final framework extension is planned to encompass intent assurance functionalities, closing essentially the intent loop. This can be achieved by incorporating a digital twin of the network to validate the expressed multimodal intent by investigating "What-if" simulation scenarios and their impact on the infrastructure prion to their deployment, without compromising the actual network's operation.


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